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Econometrics
Assignment 3**

Due Date

Friday November 7, 2025 at 11:59 PM

Directions

Answer each question clearly and concisely. Write equations using LaTeX where appropriate and explain all assumptions in your own words. Submit both a PDF and Quarto file to the nexus assignment portal.

Git and GitHub

1. (a) Create a new R project in your **econ_3201** directory called **assignment_3**.
(b) Download the assignment PDF and Quarto file the **assignment_3** folder.
(c) Commit and push the changes to your **econ_3201** repository on [GitHub.com](https://github.com).

Conceptual Foundations

1. (**5 pts**) Define the *potential outcomes framework*. Clearly explain what is meant by $Y_i(1)$, $Y_i(0)$, and D_i .

The Potential Outcome Framework is a statistical and theoretical structure used to define the causal effect of a treatment. It is based on the idea that each individual has two potential outcomes:.

$$Y_i = \begin{cases} Y_i(1), & \text{if } D_i = 1 \\ Y_i(0), & \text{if } D_i = 0 \end{cases}$$

Where $Y_i(1)$ is the outcome if the individual receives the treatment, and $Y_i(0)$ is the outcome if the individual does not receive the treatment.

D_i is a binary treatment indicator, where $D_i = 1$ if the individual is treated and $D_i = 0$, if the individual is not treated.

The observed outcome can be written as $Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$, we also cannot observe both potential outcomes for the same individual, only one.

2. (5 pts) State and explain the **Stable Unit Treatment Value Assumption (SUTVA)**. Why is this assumption critical for causal inference?

The Stable Unit Treatment Value Assumption (SUTVA) implies that the treatment received by one individual does not affect the outcomes of others, and that the treatment itself is consistent and well defined for all treated individuals.

The SUTVA assumptions is stated below:

No interference between units: Each individual's potential outcomes depend only on their own treatment status. For example, the outcome of person i should not be influenced by whether person j received the treatment.

Consistency: The treatment assigned is the same for all treated individuals, that is there are no hidden variations of the treatment.

This assumption is critical because it ensures that the potential outcomes $Y_i(1)$ and $Y_i(0)$ are well defined and comparable across individuals. Without SUTVA, it becomes impossible to isolate the causal effect of the treatment, since observed differences in outcomes might reflect interference between individuals or inconsistencies in the treatment itself.

3. (5 pts) What is the **Conditional Independence Assumption (CIA)**? Write it formally and explain its meaning in words.

According to the Conditional Independence Assumption (CIA), after accounting for the observed characteristics X_i , the process determining who receives treatment is independent of the potential outcomes. As a result, selection bias disappears.

Formally it can be written as:

$$(Y_i(1), Y_i(0)) \perp C_i | X_i$$

This expression means that, conditional on X_i , the treatment assignment C_i is statistically independent of the potential outcomes of both $Y_i(1)$ and $Y_i(0)$.

4. (5 pts) Describe the **Overlap (Common Support)** condition and its practical importance.

The condition requires that every individual has a positive probability of receiving either the treatment or not being treated

$$0 < P(D_i = 1|X_i) < 1$$

5. (5 pts) Explain why the **Average Treatment Effect on the Treated (ATT)** may differ from the **Average Treatment Effect (ATE)**.

Average Treatment Effect on the Treated(ATT): This measures the average treatment effect among individuals who actually received the treatment, that is, for those with $(D_i = 1)$. It can be expressed as:

$$ATT = E[Y_i(1) - Y_i(0)|D_i = 1]$$

Average Treatment Effect(ATE) : This measures the average treatment effect for the entire population including both treated and untreated individuals. It can be expressed as

$$ATE = E[Y_i(1)] - E[Y_i(0)]$$

Identifying Causal Effects (25 points)

1. (5 pts) Suppose treatment D_i is assigned completely at random. Show that

$$E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] = ATE$$

Explain why randomization justifies this equality.

Proof: Suppose treatment D_i is assigned completely at random. This implies that treatment assignment is independent of the potential outcomes:

$$(Y_i(1), Y_i(0)) \perp D_i$$

Because treatment is randomly assigned, the treated and control groups are statistically identical on average. Therefore, conditional expectations equal their unconditional expectations:

$$E[Y_i(1)|D_i = 1] = E[Y_i(1)] \text{ and } E[Y_i(0)|D_i = 0] = E[Y_i(0)]$$

Substituting these into the observed difference gives:

$$E[Y_i(1)] - E[Y_i(0)]$$

By definition, the Average Treatment Effect (ATE) is:

$$ATE = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)]$$

Therefore:

$$E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] = ATE$$

Randomization ensures that treatment assignment D_i is unrelated to potential outcomes $Y_i(1)$ and $Y_i(0)$. As a result, the treated and control groups are statistically equivalent both observed and unobserved characteristics are balanced across groups. Therefore, comparing the average observed outcomes of treated and untreated individuals provides an unbiased estimate of the Average Treatment Effect (ATE), free from selection bias.

2. (10 pts) Suppose treatment depends on observed covariates X_i , such that $(Y_i(1), Y_i(0)) \perp D_i | X_i$. Derive an expression for ATE using the **law of iterated expectations**, and interpret it.

Derivation

we start with the definition of the Average Treatment Effect (ATE)

$$ATE = E[Y_i(1) - Y_i(0)]$$

Applying the law of Iterated expectations condition on X_i :

$$ATE = E[E[Y_i(1) - Y_i(0)|X_i]]$$

Split the expectation:

$$ATE = E[E[Y_i(1)|X_i] - E[Y_i(0)|X_i]]$$

Using CIA to replace the unobserved counterfactuals with observed outcomes, conditional on X_i : For the treated outcome $Y_i(1)$:

$$E[Y_i(1)|X_i] = E[Y_i(1)|D_i = 1, X_i] = E[Y_i|D_i = 1]$$

For the Untreated outcome $Y_i(0)$:

$$E[Y_i(0)|X_i] = E[Y_i(0)|D_i = 0, X_i] = E[Y_i|D_i = 0]$$

Substitute the observed conditional expectations back into the ATE formulas:

$$ATE = E_X[E[Y_i|D_i = 1, X_i] - E[Y_i|D_i = 0, X_i]]$$

The expression dictates that to identify the ATE, one must first remove the selection bias by controlling for X_i , and then aggregate these conditional causal effects across the entire range of X_i observed in the population. The ATE is the weighted average of the conditional treatment effects, with weights determined by the prevalence of each X_i value in the population.

3. (10 pts) Explain how *selection bias* arises if the CIA fails. Use the decomposition:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0] = ATE + \text{Selection Bias.}$$

The Conditional Independence Assumption (CIA) states that:

$$(Y_i(1), Y_i(0)) \perp D_i | X_i$$

in words the CIA explains that once we control for the observable characteristics X_i , the treatment assignment is as good as random. However when CIA fails, the observed difference in the mean outcomes reflects both the true treatment effect and selection bias, making the estimated treatment effect biased.

Using the decomposition:

$$E[Y_i | D_i = 1] - E[Y_i | D_i = 0] = ATE + SelectionBias$$

where selection Bias

$$= E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]$$

captures the difference in what the treated and untreated groups would have achieved even without the treatment.

This implies that if people who receive the treatment are inherently different their outcomes would be higher even without the program. As a result the observed difference

$$E[Y | D = 1] - E[Y | D = 0]$$

overstate the true causal effect.

Applied Exercise in R (30 points)

Use R to simulate data, estimate treatment effects, and interpret your results. Include all R code and output.

```
set.seed(123)

n <- 1000
X <- rnorm(n)
D <- ifelse(X + rnorm(n) > 0, 1, 0)
Y0 <- 2 + 0.5 * X + rnorm(n)
Y1 <- Y0 + 3 + 0.5 * X
Y <- ifelse(D == 1, Y1, Y0)
data <- data.frame(Y, D, X)
```

1. (5 pts) Compute the difference in average outcomes between treated and untreated. Interpret the result in terms of potential bias.

```
Treated <- mean(data$Y[data$D == 1])
Untreated <- mean(data$Y[data$D == 0])
Naive_Difference <- Treated - Untreated
Naive_Difference
```

```
[1] 3.738236
```

Result Interpretation: The given Average Treatment Effect (ATE) $E[Y_1 - Y_0] = E[(Y_0 + 3 + 0.5 * X) - Y_0] = 3$. The observed difference in average outcomes between the treated and untreated is approximately 3.7, which is an overestimate of the true ATE. This occurs because treatment assignment depends on X and individuals with higher X values are both more likely to receive treatment and have higher outcomes even without treatment. Therefore, the difference in means captures not only the causal effect of treatment but also the positive selection bias arising from the correlation between D_i and X_i .

2. (5 pts)

(a) Estimate the treatment effect controlling for (X):

$$Y_i = \alpha + \tau D_i + \beta X_i + \varepsilon_i$$

```
ols <- lm(Y ~ D + X, data = data)
summary(ols)
```

Call:

```
lm(formula = Y ~ D + X, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.8775	-0.6586	-0.0137	0.6656	3.0932

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.16582	0.05072	42.70	<2e-16 ***
D	2.90330	0.07586	38.27	<2e-16 ***
X	0.76994	0.03822	20.15	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.003 on 997 degrees of freedom

Multiple R-squared: 0.795, Adjusted R-squared: 0.7946

F-statistic: 1933 on 2 and 997 DF, p-value: < 2.2e-16

(b) Compare the estimate of (τ) to the true treatment effect of 3.

The OLS estimate $(\hat{\tau}) = 2.903$, is very close to the true treatment effect of 3. The coefficient on D_i represents the causal impact of the treatment after accounting for the observed covariate X_i . By including X_i in the regression, we control for confounding factors that influence both treatment assignment and outcomes, thereby removing most of the bias present in the naive difference (3.7)

3. (5 pts) Propensity Score Matching

(a) Estimate treatment effects using matching. Search ?matchIt in the console

```
library('MatchIt')
Matching_Model<- matchit(D ~ X, data = data, method = "nearest")
```

Warning: Fewer control units than treated units; not all treated units will get a match.

```
summary(Matching_Model)
```

Call:

```
matchit(formula = D ~ X, data = data, method = "nearest")
```

Summary of Balance for All Data:

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio	eCDF Mean
distance	0.6735	0.3623	1.3664	0.9226	0.3214
X	0.5301	-0.5543	1.2878	1.0580	0.3214

	eCDF Max
distance	0.4986
X	0.4986

Summary of Balance for Matched Data:

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio	eCDF Mean
distance	0.7226	0.3623	1.5822	0.5775	0.3719
X	0.6856	-0.5543	1.4724	0.7930	0.3719

	eCDF Max	Std. Pair Dist.
distance	0.5802	1.5822
X	0.5802	1.4724

Sample Sizes:

	Control	Treated
All	474	526
Matched	474	474
Unmatched	0	52
Discarded	0	0

```
matched_data <- match.data(Matching_Model)
att_Matching_Model <- with(matched_data, mean(Y[D == 1]) - mean(Y[D == 0]))
att_Matching_Model <- lm(Y~D , data = matched_data)
summary(att_Matching_Model)
```

Call:

```
lm(formula = Y ~ D, data = matched_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.4847	-0.7626	0.0121	0.7418	3.9291

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.73907	0.05245	33.15	<2e-16 ***
D	3.88024	0.07418	52.31	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.142 on 946 degrees of freedom

Multiple R-squared: 0.7431, Adjusted R-squared: 0.7428

F-statistic: 2736 on 1 and 946 DF, p-value: < 2.2e-16

(b) Compare the matched estimate to OLS and the naïve difference.

```
Naive_Difference <- mean(data$Y[data$D == 1]) - mean(data$Y[data$D == 0])

ols <- lm(Y ~ D + X, data = data)
ols_Estimate <- coef(ols)["D"]

matched_data <- match.data(Matching_Model)
att_coefficient <- coef(att_Matching_Model)["D"]

Comparison <- c(Naive = Naive_Difference,
```



```

        OLS = ols_Estimate,
        ATT_Matched = att_coefficient)

print(Comparison)

```

Naive	OLS.D	ATT_Matched.D
3.738236	2.903303	3.880237

4. (5 pts) Simulate a violation of the Conditional Independence Assumption by introducing an unobserved confounder (U_i):

```

U <- rnorm(n)
D <- ifelse(X + U + rnorm(n) > 0, 1, 0)
Y0 <- 2 + 0.5 * X + 0.5 * U + rnorm(n)
Y1 <- Y0 + 3 + 0.5 * X
Y <- ifelse(D == 1, Y1, Y0)
bias_model <- lm(Y ~ D + X, data = data.frame(Y, D, X))
summary(bias_model)

```

Call:

```
lm(formula = Y ~ D + X, data = data.frame(Y, D, X))
```

Residuals:

Min	1Q	Median	3Q	Max
-4.0005	-0.7182	0.0067	0.7331	4.0364

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.80830	0.05294	34.15	<2e-16 ***
D	3.67837	0.07941	46.32	<2e-16 ***
X	0.56640	0.04006	14.14	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.104 on 997 degrees of freedom

Multiple R-squared: 0.7939, Adjusted R-squared: 0.7934

F-statistic: 1920 on 2 and 997 DF, p-value: < 2.2e-16

Visualize and discuss how this unobserved confounder biases the estimated treatment effect.

Because U is unobserved in the regression, OLS can't control for its influence. As a result, the estimated coefficient on $D = 3.65$ ends up capturing not only the true treatment effect but also part of U effect. This means the OLS estimate is biased upward compared to the true causal effect of 3. If the Conditional Independence Assumption (CIA) is violated due to an unobserved confounder, the OLS estimate won't reflect the true causal effect.

```
plot(X, Y, col = ifelse(D==1, "red", "blue"), pch = 16,
     main = "Outcome vs Covariate with Confounding",
     xlab = "X", ylab = "Y")
legend("topleft", legend = c("Treated", "Untreated"),
     col = c("red", "blue"), pch = 16)
```

